Job Loss and Occupational Job Security: Evidence from Post-Displacement Sorting

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Abstract

While previous research has extensively studied earnings losses after job displacement, relatively little is known about whether displaced workers move into jobs that are inherently less secure, potentially contributing to the persistent costs of job loss. This study estimates the effect of displacement on the job security associated with the occupations into which displaced workers sort following dismissal. Using a dataset containing the employment histories of approximately four million individuals working in Italy, which allows me to disentangle voluntary from involuntary job moves, I construct two sector-specific occupational indicators of job security that I attach to each employment spell: one capturing the risk of unemployment and the other measuring expected tenure. Then, leveraging collective dismissals as a source of exogenous variation and their staggered treatment timing, I find that displacement raises the risk of unemployment inherent to post-displacement occupations by about 2.36 percentage points and reduces expected tenure by around 155 days, on average across post-treatment periods. These effects correspond to declines in occupational job security of approximately 13 and 12 percent, respectively, relative to pre-displacement averages.

JEL codes: J63, J62, J24.

Keywords: Job displacement, Occupational sorting, Job security, Unemployment scarring, Job ladder.

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1 Introduction

Job displacement has persistent adverse effects on workers' careers. Since Heckman and Borjas (1980), who first identified the methodological challenges involved in estimating the impact of job loss on workers' labor-market outcomes, an extensive body of literature has emerged documenting displacement's effects on earnings (Jacobson et al., 1993), wages (Ruhm, 1991; Arulampalam, 2001), and subsequent unemployment risk (Gregg, 2001). Yet, one aspect that remains underexplored is how job sorting following displacement affects subsequent occupational outcomes, which may ultimately contribute to the recurring unemployment spells observed over the course of displaced workers' careers.

This study is motivated by the premise that displaced workers' heightened risk of recurring unemployment may be closely linked to the occupations that they find after losing their jobs. In particular, displacement is a disruptive event that may confine workers to a labor market with limited opportunities, where they can only compete for lower-quality occupations. This may occur, for instance, because more attractive vacancies open up less frequently and take longer to fill, making them de facto unattainable for recently displaced workers with poor outside options. If the set of occupations accessible to displaced workers tends to include the generally less desirable ones, then these occupations may not only offer lower compensation but also come with reduced job security. Consequently, sorting into these occupations may, in and of itself, constitute a fundamental driver of the persistent adverse effects of displacement.

The proposed analysis tests the hypothesis that displacement leads workers to sort into occupations with lower job security than those they held before displacement. In other words, the objective is to assess whether displacement induces workers to move down the *occupational job-security ladder*. To test this hypothesis, I use a dataset containing the employment histories of approximately four million individuals working in Italy. A key feature of this dataset is the availability of information on the reason for each job termination, which allows me to disentangle voluntary from involuntary job moves. This information was crucial for developing two job security indicators, which I assign to each sector-specific occupation in the sample (hereafter loosely referred to simply as "occupations"): one capturing the risk of unemployment and the other measuring expected tenure.

Several factors may account for differences in job security across occupations. First, unions may have a stronger presence in some sectors and occupations and may be more effective at protecting incumbent workers than new hires, making entry-level positions more vulnerable to dismissal. These occupations may also be subject to less stringent employment protection laws. Second, some occupations may be more exposed to the risks of automation or offshoring, or more sensitive to business-cycle fluctuations. Third, some sectors, by their nature, tend to have a higher prevalence of seasonal or temporary work. The job security measures developed for this analysis are designed to capture this rich heterogeneity, which is the central focus of this study.

I estimate the effects of displacement on occupational job security using a difference-in-differences analysis that leverages collective dismissals as a source of exogenous variation and their staggered treatment timing to define treatment and control groups. The treatment group then consists of workers who have already experienced a collective dismissal, while the control group includes those who will experience one at a later stage. The motivation for using later-treated individuals, rather than never-treated ones, as a control group rests on the assumption that workers subject to collective dismissals, albeit at different points in time, are more likely to share similar unobservable characteristics with one another than with those who were never treated, making them more suitable comparison units. Nonetheless, I also conducted a supplementary analysis using a more conventional approach that relies on never-treated individuals as controls. The resulting estimates are used as a benchmark for the main specification. To the best of my knowledge, the proposed approach to estimating the costs of job loss is novel in the literature. Moreover, using collective dismissals offers an important advantage over existing designs that rely on mass layoffs as exogenous shocks. While both types of events are firm-level workforce adjustments motivated by economic or organizational reasons (and are therefore plausibly orthogonal to dismissed workers' unobservable abilities) collective dismissals are generally smaller-scale shocks than mass layoffs.[1] As a result, concerns about substantial spillover effects in local labor markets, often raised in the context of mass layoffs (e.g., Cederlöf, 2024), are considerably mitigated.

The main findings indicate that displacement leads to a significant decline in job security associated with the occupations found by displaced workers. In particular, these occupations feature a significantly higher inherent risk of unemployment of approximately 2.36 percentage points and a reduced expected tenure of about 155 days. To give a sense of the magnitude of these effects, the increase in unemployment risk corresponds to a 12.8% rise relative to the average value before displacement, while the drop in expected tenure represents an approximately 11.7% decline. Notably, these effects seem to be relatively stable and persistent over time.

Moreover, examining the heterogeneity of these findings across different subgroups, such as gender, level of education, and geographical region of employment (North, Center, and South of Italy), I find estimates that suggest a larger decline in occupational job security for women and for individuals working in the northern regions of Italy. On the other hand, estimates for workers with different levels of educational attainment do not provide a clear-cut picture, although the negative effects of displacement appear more persistent among lower-educated individuals. Nevertheless, wide confidence intervals do not allow me to claim statistically significant gaps among any of these subgroups.

Examining these effects in more detail, by inspecting average post-treatment effects (defined as the mean of the estimated coefficients across all post-treatment periods), I find that men experience a 2.1 pp increase in unemployment risk, while women seem to endure more severe consequences of

1 Section 2.1 provides an extensive discussion of collective dismissal procedures. In short, these procedures, within the Italian legislation, apply to companies with more than 15 employees that intend to dismiss at least five employees. This implies, for instance, that if a company of 50 employees wants to dismiss 10% of its workforce (i.e., five employees), this would qualify as a collective dismissal. On the other hand, the literature (see, for example, Bertheau et al., 2023) usually defines mass layoffs as a 30% reduction in a firm's workforce within a year, typically restricting the focus to sufficiently large firms. Hence, collective dismissals can also entail firm-level shocks that are smaller in scale than mass layoffs.

displacement, with an increase of 2.72 pp. The decline in expected tenure is in line with these results, with men and women facing reductions of approximately 149 and 163 days, respectively. Turning to the dynamics of these effects, the decline in occupational job security remains relatively persistent over time for women, whereas men exhibit a gradual recovery. Differences between higher-educated (those holding at least a college degree) and lower-educated individuals, instead, seem to emerge only in later periods and are clearly discernible only for the expected tenure indicator. On average, the expected tenure of lower-educated individuals decreases by about 168 days following displacement, compared with a more moderate reduction of approximately 97 days among higher-educated workers. This marked difference is driven almost entirely by the recovery observed among higher-educated individuals in later periods. Finally, workers employed in the North of Italy at the time of dismissal seem to be more adversely affected by displacement than workers in other regions, experiencing an increase in unemployment risk of about 2.53 percentage points and a reduction in expected tenure of approximately 179 days. These effects also appear to be more persistent for workers in the North. While these results may initially seem counterintuitive, they are in fact consistent with the premise that more advantageous initial conditions entail greater potential losses. Thus, although workers in northern Italy might be expected to transition back into employment more quickly, their higher levels of pre-displacement occupational job security also imply larger subsequent declines relative to individuals in central and southern Italy, who were employed in comparatively less secure occupations before displacement.[2]

These findings are robust to a comprehensive set of robustness, sensitivity, and placebo checks addressing several potential concerns. First, I assess the accuracy of the constructed indicators, given that sectors and occupations in the sample differ in their level of representation. Second, I examine whether including collective dismissals as involuntary job-loss events when constructing the indicators may raise meaningful endogeneity concerns. Third, I assess whether the estimates are biased by spillover effects at the regional level. Fourth, I test whether the results are driven by a small number of outlier observations. Fifth, since the baseline analysis focuses on workers aged 30 to 54 (as workers at the beginning or end of their careers may react differently to displacement than prime-age workers), I extend the sample to include all individuals aged 20 to 64. Finally, to verify that a significant change occurs around the time of treatment, I arbitrarily reassign the treatment three years before its actual occurrence. The first five exercises yield estimates that are close to those from the main specification, while the placebo test confirms that the estimated effects reflect a fundamental change occurring around the displacement year.

This study contributes to the literature on the costs of job loss. While previous research has extensively documented the adverse consequences of displacement on workers' career trajectories, particularly in terms of earnings losses,[3] this analysis directs its focus toward the search and matching

2 These results are consistent, for instance, with the evidence presented in Fackler et al. (2021), which documents that wage losses increase substantially with pre-displacement employer size, reflecting the loss of large-firm wage premiums.

3 A non-exhaustive list of contributions includes: Ruhm (1991); Jacobson et al. (1993); Stevens (1997); Arulampalam (2001); Gregg (2001); Burgess et al. (2003); Eliason and Storrie (2006); Von Wachter and Bender (2006); Oreopoulos et al. (2012); Lachowska et al. (2020); Schmieder et al. (2023).

outcomes of displaced workers. The central idea advanced in this study is that displacement may constrain workers' options, leading them to accept lower-quality occupations that inherently entail reduced job security. Huckfeldt (2022) shares a similar conceptual framework, developing a model in which firms hire more selectively during recessions, inducing displaced workers to optimally search for reemployment in lower-paid jobs. That analysis finds that the earnings costs of job loss are borne almost entirely by workers who switch occupations following displacement, underscoring the importance of occupational mobility in shaping the negative effects of job loss. In contrast, I remain agnostic about the underlying mechanisms that constrain displaced workers' options and induce them to switch occupations. Furthermore, rather than focusing on earnings losses, I estimate the effects of displacement on occupational job security, assessing the extent to which displaced workers move down the occupational job-security ladder. This paper is also related to Figueiredo et al. (2026), which estimates the effects of displacement on job security by examining whether displaced workers sort into either permanent or temporary jobs. I take a different approach from this paper by classifying occupations according to their intrinsic level of job security and identifying them as a key channel through which displaced workers face heterogeneous exposure to the risk of dismissal.

More broadly, this study also speaks to the job ladder literature (Burdett and Mortensen, 1998), which posits that displacement may disrupt workers' progress toward higher-paying and more secure positions, pushing them into less secure, lower-paid jobs (Pinheiro and Visschers, 2015; Jarosch, 2023).[4] Being displaced thus entails moving down the job ladder into positions associated with a higher probability of layoff. This paper contributes to this literature both empirically and methodologically by quantifying the decline in occupational job security following displacement. As noted in Pinheiro and Visschers (2015), unemployment scarring may arise even when labor supply is assumed to be homogeneous, implying that demand-side factors alone could, in principle, generate persistent adverse effects of job loss. The research design adopted in this study moves in this direction by exploiting the staggered timing of collective dismissals as exogenous shocks and thereby minimizing unobserved worker heterogeneity between the treatment and control groups. This empirical strategy helps isolate demand-side mechanisms from supply-side selection, strengthening the identification of the causal effects of job loss.

This paper is structured as follows: Section 2 describes the data and the institutional setting, Section 3 outlines the two measures of occupational job security, Section 4 motivates the empirical strategy, the results are presented in Section 5, followed by robustness checks and placebo in Section 6, and Section 7 concludes.

4 The positive correlation between job security and wages has been widely discussed in the literature on the lack of compensating differentials (Mayo and Murray, 1991; Winter-Ebmer, 2001; Bonhomme and Jolivet, 2009). For example, Mayo and Murray (1991) shows that workers sort into large or small firms based on their unobservable quality. In this context, smaller firms, which typically offer both lower wages and less stable employment, may attract workers with generally more unstable labor-market histories, who are willing to accept worse employment conditions given their poor outside options. Unlike these works, I do not focus on firm-level outcomes but rather on the sectoral-occupational level. However, Haltiwanger et al. (2022) shows that most of the wage dispersion (and plausibly most of the job security dispersion) occurs at the sectoral level, which supports the relevance of the indicators used in this study.

2 Data and institutional setting

This study relies on a sample extracted from the "Comunicazioni Obbligatorie" database,[5] which records mandatory notifications that Italian employers (or their intermediaries) are required to submit to the competent regional or national authority regarding worker flows, such as hires, separations, or contract conversions.[6] The sample comprises around 22 million observations, drawn from a random selection of 4 million individuals employed in the Italian labor market, in both the private and public sectors, from 2010 to 2022.[7] The data contains the complete employment histories of the selected individuals throughout the period considered. In its original structure, each row reports information on a single employment spell, which includes the starting and ending dates, the contract type (temporary or permanent, full-time or part-time), the occupation, the sector of employment, and notably, the reason for job termination, which is essential for the computation of the outcome variables. The dataset also includes several time-invariant individual characteristics such as gender, year and region of birth, and education.[8] Although the data contains an income variable, it is recorded only at the time of hiring and only for spells starting after 2010, making it unsuitable for earnings analysis.

To prepare the dataset for the analysis, I restructured it into a yearly panel format, in which each year records each individual's main employment.[9] Next, I extracted the sample of interest, which consists of individuals who experienced only one collective dismissal during the sample period. I limited the analysis to these cases to avoid issues in interpreting the estimates that may arise in the presence of multiple treatment episodes.[10] Additionally, I excluded workers who started a new job before their prior contract's official termination with the dismissing firm, who were dismissed from a job that was not identified as their main one, or who returned to the same firm that had previously dismissed them. This last cut avoids retaining cases of worker reinstatement when a collective dismissal is later ruled illegitimate by a court. To further homogenize the sample, I restricted it to individuals aged at least 30 in 2010 and not older than 54 in 2022. This choice was motivated by the fact that young and senior workers may respond differently to displacement than prime-age workers: younger individuals frequently change jobs early in their careers regardless of dismissal, while older

5 The full name of the dataset is "Campione Integrato delle Comunicazioni Obbligatorie".

6 Note that individuals appear in the dataset only if at least one of these events occurs within the sample period.

7 The random selection is performed by extracting every individual born on the 1st, the 9th, the 10th, and the 11th day of any month and any year.

8 The reported level of education in the dataset refers to the one at the time of the last registered employment spell. This is a non-trivial caveat, as it is not possible to know whether individuals upgraded their education once they started working.

9 Similarly to Card et al. (2013), I restricted the sample to keep a single dominant record per year. However, as earnings are not the focus of this study, I defined the main job as the one with the longest employment spell within each year. To identify main jobs, I sequentially retained the employment record that displayed: 1. the highest number of employment days within the year; 2. the longest employment spell overall (spanning throughout the whole sample period); 3. full-time jobs were preferred to part-time ones; 4. the highest income (however, the income variable is imprecise, as it is reported only at the time of hiring and for employment spells that began after 2010). When two or more employment spells were found to be equal in every one of these categories, I picked one at random.

10 However, including individuals who were treated more than once does not appreciably change the results, see Figure B1. This may be due to the relatively low number of individuals who experienced a collective dismissal more than once in the sample period (only 1.7% of the total).

workers may instead react to displacement by anticipating their retirement.[11] Finally, to obtain a balanced panel, I also excluded individuals who had gaps in their reported employment history (i.e., a full calendar year without a single employment day), which effectively removes the long-term unemployed from the sample. The main rationale for excluding these workers is that I cannot assign them a measure of job security for the years in which they were not employed. Moreover, workers who remain unemployed for an extended period may carry unobservable characteristics that fundamentally differentiate them from workers with more stable employment histories, potentially introducing significant confounders into the analysis.[12] This changes the interpretation of the results, as the effect of displacement on occupational job security is estimated only conditional on workers reporting at least one employment spell in each year.[13] After these adjustments, and after excluding individuals who experienced a collective dismissal within the first three years of the period considered (to allow for at least three pre-treatment years), the final sample consists of 8,256 treated individuals observed over 13 years, corresponding to a total of 107,328 observations.[14] Summary statistics for this sample are shown in Appendix A from Table A1 through Table A6. The size of all treatment cohorts is sufficiently large, although earlier cohorts, from 2013 to 2017, are considerably larger than later ones. The distribution of individuals across treatment cohorts is reported in Table A7 in Appendix A. The main specification will only make use of this group of units, exploiting later-treated units to build control groups that vary by cohort and calendar time.

Nevertheless, I replicate the analysis using the conventional approach in which never-treated units serve as the control group. In this case, controls are typically selected based on observable characteristics that make them suitable comparison units for extrapolating untreated potential outcomes. Since the estimator used in the analysis, proposed by Callaway and Sant'Anna (2021), already incorporates a propensity score matching procedure, it is not necessary to identify suitable control units prior to the estimation stage. When never-treated units are included, the resulting sample comprises 3,364,361 observations and 258,797 individuals, of whom 7,449 are treated.[15] In the following sections of this paper, I will mainly focus on estimates obtained with the sample that includes treated units

11 For example, Farber et al. (2019), who studies the role of employment and unemployment histories in callbacks to job applications, finds that younger and older applicants have a lower callback rate than prime-aged applicants. Furthermore, also in Bertheau et al. (2023), only workers who are at most 50 years old in the year preceding displacement are retained to limit the influence of early retirement programs. In contrast to this approach, however, I preferred to further homogenize the age composition of the sample, given that later-treated units are used in this work to construct counterfactual potential outcomes and thus might generally be slightly older than already-treated units. In Bertheau et al. (2023), instead, the authors employ a more standard identification strategy that exploits never-treated units as controls.

12 Eriksson and Rooth (2014), for example, finds that potential employers attach a stigma to contemporary unemployment spells that last at least nine months, which negatively affects their hiring decisions of long-term unemployed individuals.

13 Other papers in the literature, such as Ruhm (1991) and Huckfeldt (2022), adopt a similar approach. The former paper considers only "individuals with fairly strong attachments to the labor force," while the latter focuses only on workers who were employed full-time in both their previous and post-displacement jobs, thereby excluding workers who did not find reemployment.

14 As opposed to what is common practice in the literature on the cost of displacement (e.g., see Jacobson et al., 1993), the sample was not restricted to include only individuals with at least three years of tenure in the firing firm at the time of dismissal. This choice was motivated by the fact that outcome variables, described in Section 3, are time-invariant indicators of job security and, thus, only vary when workers change employment. Hence, keeping only individuals who have not changed firms over the three years preceding the collective dismissal would have implied ending up with a complete absence of variation in the outcome variable for those periods, which would have made these pre-treatment coefficients impossible to compute.

15 Note that the number of treated units decreases relative to the previous specification because, to homogenize the number of pre- and post-treatment periods, I exclude all units that are treated during the last three years of the sample.

only. The motivation for this choice will be outlined in Section 4. Nonetheless, as identification strategies that employ never-treated units have been the standard, the charts and the tables showing the results for the specifications that use the untreated-units sample are reported in Appendix B, serving as a benchmark for the main results.

2.1 The Italian institutional setting of collective dismissals

The collective dismissal (*licenziamento collettivo*) procedure in Italy is the process through which a company can dismiss a number of employees for economic, organizational, or structural reasons.[16] In the dataset that I employ for this analysis, 0.62% of all job separations are filed as collective dismissals. These procedures apply to companies with more than 15 employees that intend to dismiss at least five employees within 120 days, and they involve several steps.[17] First, the company must send a written notification to the trade unions and the relevant labor authorities, providing detailed information about the reasons for the dismissals, the number and roles of employees affected, and the planned timeline. Then, within seven days, trade unions may propose to re-examine the reasons for the collective dismissal and discuss possible alternative solutions with the company, such as temporary layoffs, retraining, or reduced working hours. This consultation phase lasts a maximum of 45 days (or about 22 days if the procedure involves fewer than ten employees). If no agreement is reached by the deadline, administrative authorities may initiate a second consultation phase, which must end within 30 days (15 if the procedure involves fewer than ten employees). Once both consultation phases are over, the company can proceed to dismiss the workers that it deems redundant. To do so, the company must apply criteria that had been previously determined with union agreements. Typically, these criteria include length of service (newcomers shall leave the company first, similar to the "last-in, first-out" rule found in the labor-market legislation of other European countries), the presence or absence of family responsibilities, and the company's operational needs. Finally, dismissed employees are entitled to severance pay and may qualify for unemployment benefits.

16 Collective dismissal procedures are regulated by Law 223/1991 and subsequently refined by Legislative Decree 151/1997, both implementing European directives aimed at harmonizing legislation on collective redundancies across EU member states.

17 One potential concern is the possible presence of anticipation effects. If workers are notified of their imminent dismissal, they may begin searching for a new job before the actual displacement date. To address this issue, I examine whether including workers who start their subsequent job within a window of 120 days before their actual dismissal affects the results. While effects are slightly diminished, including these workers does not change the estimates substantively. The corresponding results are reported in Figure B2 in Appendix B.

3 Measuring occupations' job security

I construct two indicators of job security associated with each sector-specific occupation[18] in the sample: one assessing the risk of unemployment and the other measuring expected tenure. These indicators were calculated for 2131 sectors and 531 occupations separately, and then averaged to derive a single measure of job security for each sector-specific occupation (hereafter loosely referred to as "occupations" for the sake of simplicity). This process yields an indicator of job security for 9316 distinct occupations present in the final dataset of treated units.

To construct the two job security measures, I followed slightly different procedures. For the unemployment risk indicator, I exploited a crucial piece of information in the dataset, namely the reason for each job termination. Then, observing the annual survival rates of each job, I categorized termination events, assigning a value of one to involuntary separations and a value of zero to voluntary job transitions and ongoing employment spells.[19] Subsequently, I computed the share of involuntary separations for each occupation and sector separately, which I then averaged to obtain a time-invariant measure that captures the risk of unemployment attached to any given sector-specific occupation. This process is summarized by the formula below:

$$y_{jk} = \frac{\left(\frac{\sum L_{itj}}{N_j} + \frac{\sum L_{itk}}{N_k}\right)}{2}$$

where y is the unemployment risk indicator pertaining to each occupation, defined as the combination of sector j and occupation k of individual i, L is a dummy variable that takes the value zero for voluntary job moves and ongoing work spells within year t or the value of one when the worker is subject to a layoff, and N is the number of observations for any given sector and occupation.

I developed the second indicator, which measures expected tenure, in a similar fashion. However, unlike the previous case, constructing this indicator did not require inspecting jobs' annual survival rates, as it was sufficient to calculate their length using the hiring and separation dates provided in the data.[20] Nevertheless, information on the reasons for separation was still useful, as it allowed me to exclude from the computation employment spells that ended with voluntary job moves.[21] Once I

18 Sectors refer to 6-digit ATECO codes, while occupations are defined as "the set of activities that an individual must perform in carrying out their job" (as defined by the Italian National Institute of Statistics, according to the classification CP2011). These activities imply knowledge and skills and are codified and regulated.

19 To conduct this operation, employment spells that reported a termination reason that left unclear the categorization into voluntary or involuntary separations, such as firm closures, were discarded. In the case of firm closures, for example, it was not possible to determine whether the firm closure entailed an actual shutdown, with all its employees being made redundant, or whether the enterprise was acquired by another company, resulting only in a transfer of employment from one employer to another. Discarding all separations whose voluntary or involuntary nature could not be determined removes about 4% of the employment spells used to compute the job security indicators.

20 Note that the start date of employment spells observed in the dataset may precede 2010, the first year in the panel. Since the dataset records the original start date for all employment spells, the expected tenure indicator can be computed accurately regardless of when the spell began. Consequently, left-truncation does not affect the construction of this measure. However, right-censoring arises because employment spells may extend beyond the observation period. By contrast, the unemployment risk indicator is computed as an annual occupation-specific survival rate. Hence, its construction is not affected by either left-truncation or right-censoring.

21 Employment spells with unclear separation reasons, as mentioned before, were also excluded.

obtained these mean values for each occupation and sector, I averaged these employment durations to obtain a single measure of expected tenure attached to each sector-specific occupation, as illustrated before for the other indicator. The formula used to compute this second measure closely resembles the previous one:

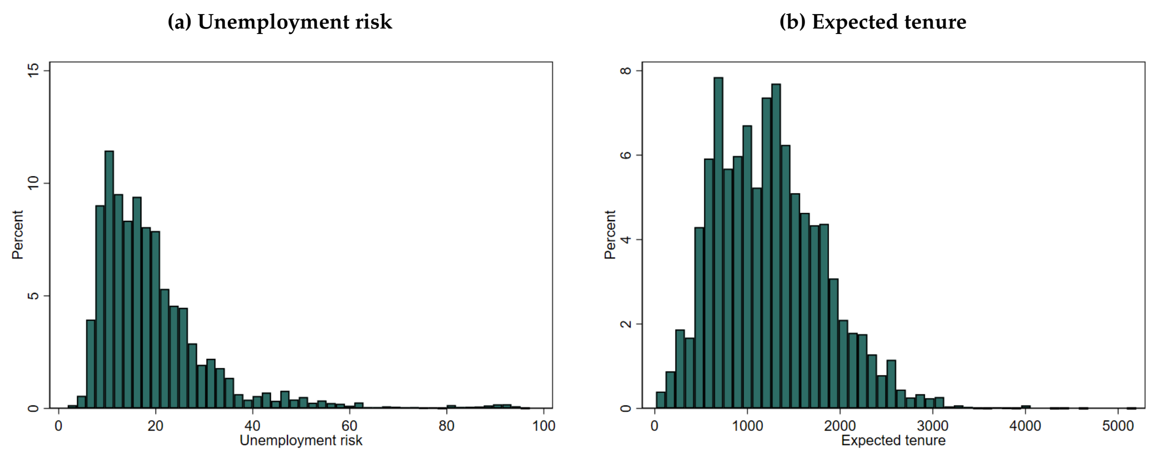
$$y_{jk} = \frac{\left(\frac{\sum d_{ij}}{N_j} + \frac{\sum d_{ik}}{N_k}\right)}{2}$$

where y is now the expected tenure indicator for each occupation, while d denotes the number of days worker i spent working in that occupation.

Both indicators should reflect the heterogeneity in job security associated with being employed in different occupations.[22] Finally, it is important to note that both measures of job security are time-invariant and thus convey the level of job security for each occupation for the whole sample period. Since the indicators are time-invariant, workers who do not switch occupations exhibit no change in occupational risk.

Table 1 shows mean values of both outcome variables before and after treatment, while Figure 1 displays the distribution of the two outcome variables' values in the final sample, regardless of the treatment period.[23]

Figure 1: Distribution of the outcome variables' values in the sample of treated units



Notes: The two charts show the distribution of the two outcome variables' values in the final sample of treated units. The charts, therefore, display values belonging to both pre- and post-treatment periods. The figures refer to the 8256 individuals in the final sample of treated units.

22 To make these measures easier to grasp, some examples of occupations characterized by high job security are professionals who work in the banking sector and engineers/designers who work in the automotive or aircraft production sector, whereas examples of low job security occupations are day laborers who work in the farming sector and professionals who work in the film industry or theaters.

23 See Figure B3 in Appendix B to compare with pre and post-treatment period figures.

Table 1: Mean value of the outcome variables before and after the treatment's occurrence

	Unemployment risk	Expected tenure
Before displacement	18.43	1330.47
	(11.19)	(600.18)
After displacement	20.46	1166.37
	(13.53)	(564.69)

Notes: Standard errors in parentheses. The table displays the average value for the two outcome variables (unemployment risk and expected tenure) before and after displacement takes place. The numbers refer to the 8256 individuals in the final sample of treated units.

3.1 Validation of occupational job security measures

To evaluate whether the proposed job security measures reasonably reflect the likelihood of dismissal intrinsic to each occupation, I categorized the sample of workers dismissed through a collective dismissal into two groups: those who sorted into a new occupation with higher job security than their pre-displacement one, and those who found an occupation with lower or equal job security. Then, I computed workers' nonemployment days for each year following the first one in which they found their new jobs and estimated the following regression:

$$y_{it} = \beta_1 D_i + \beta_2 X_i + \delta_t + \varepsilon_{it} \tag{1}$$

where y are worker i's nonemployment days in year t, D is a dummy variable that takes value 1 if the worker found a worse occupation in terms of job security following the collective dismissal, and zero otherwise, X is a vector of workers' time-invariant characteristics (age, gender, level of education), and δ are year dummies.

Table 2: Difference in days of nonemployment among workers who found a lower-job-security occupation after dismissal versus workers who found a higher-job-security occupation

	Nonemployment days	
Variables	Unemployment risk	Expected tenure
Worse outcome after treat	4.46***	3.12***
	(0.50)	(0.50)
Time-invariant observables	✓	✓
Year dummies	✓	✓
Constant	24.06***	24.46***
	(3.65)	(3.66)
Observations	50,561	50,561
R-squared	0.0145	0.0137

Standard errors are reported in parentheses below the coefficients.
Stars indicate p-values, with: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

Table 2 shows the results of this exercise, indicating that workers who found an occupation featuring a worse unemployment risk (*expected tenure*) than their pre-displacement occupation suffer, on average, 4.5 (*3.1*) more nonemployment days than those who sort into a more secure occupation. In relative terms, this corresponds to approximately 31% (*21%*) more nonemployment days. I interpret this result as descriptive evidence that the proposed indicators successfully provide a reasonably accurate measure of the intrinsic probability of involuntary dismissal.

4 Identification and Empirical Strategy

The aim of this analysis is to assess the impact of displacement on occupational job security. This task poses substantial methodological challenges, as dismissals may be correlated with unobservable characteristics of workers that could influence their future job prospects. Partialling out these features is therefore essential to obtaining causal estimates of the effects of displacement on workers' labor-market outcomes. To tackle this issue, researchers have typically resorted to firm-level exogenous shocks, such as mass layoffs and firm closures. These shocks are considered exogenous to workers' characteristics, as firms do not select the pool of workers to be dismissed based on their performance but rather according to economic and organizational needs. Hence, in the context of mass layoffs, all types of workers may be exposed to displacement risk, irrespective of their individual attributes.[24] However, Cederlöf (2024) highlights a potential limitation of this approach, noting that mass layoffs may generate spillover effects at the local or sectoral level that could amplify the estimated negative effects of displacement. This concern is clearly more relevant when mass layoffs represent a sizeable portion of the local or sectoral labor market. Studies that employ mass layoffs as sources of exogenous variation typically define them as a reduction of at least 30% of a firm's workforce within a given year. Nevertheless, these works usually rely on data lacking information on the reason for employment termination, which forces researchers to use large workforce reductions to distinguish job separations due to economic reasons from individual layoffs.[25] This study, instead, uses a dataset that includes this information, allowing me to distinguish between these two separation reasons. Furthermore, while collective dismissals retain the character of firm-level shocks motivated by economic or organizational reasons, they are generally smaller in scale than mass layoffs, alleviating concerns that spillover effects may be too pronounced.

The following analysis employs a difference-in-differences event study design in which the control group consists of workers who have not yet been displaced through a collective dismissal. The motivation for choosing not-yet-treated units as a control group lies in the idea that workers who undergo a collective dismissal, even at different points in time, are more likely to share similar unobservable traits with one another than with those who were never treated. Consequently, estimates

24 The seminal contribution in this literature is Jacobson et al. (1993), which was followed by a large body of work using the same or closely related empirical strategies.

25 This is, for instance, explicitly outlined in Bertheau et al. (2023), where to correctly specify the treatment group, they must impose a 30% annual drop in a given firm's workforce in order to avoid mischaracterizing voluntary separations as layoffs.

obtained using not-yet-treated units to derive counterfactual potential outcomes can be presumed less vulnerable to biases stemming from unobservable time-varying factors, as such confounders are more likely to be implicitly netted out. In the next section, I will therefore only present results obtained from regressions that exploit not-yet-treated units as a comparison group, but to benchmark these estimates, I also carried out the analysis using never-treated individuals as control units. These results will be reported in Appendix B.

The methodology chosen to obtain the event-study estimates is the estimator proposed by Callaway and Sant'Anna (2021) (henceforth CS). This methodology is particularly well suited to this empirical framework for two reasons. First, as mentioned above, it is robust to heterogeneous treatment effects, which is crucial in this setting, given that workers in the sample are employed in diverse occupations and face dismissals at different points in time, thereby encountering varying labor-market conditions and regulations. Second, by leveraging the "no anticipation" assumption, this estimator maximizes the use of not-yet-treated units in the control group, drawing on every cohort that has not yet been treated. This feature of the CS estimator is particularly valuable given that the dataset includes fewer treated units in later periods. It should be noted that the CS estimator employs covariates to generate a propensity score that matches units on a defined set of observable characteristics fixed at their values in the year prior to treatment. The parallel trends assumption required for identification is then the following: conditional on a set of observable characteristics, treated and control units followed similar occupational job security paths prior to dismissal, and in the counterfactual scenario in which displacement had not occurred, the distance between these trajectories would have remained unchanged. Given this conceptual framework, the main specification is:

$$y_{itg} = \sum_{e,\, e\neq -1} \beta_e \times D_i \times \mathbb{1}(t = g + e) + \pi X_i + \varepsilon_{itg} \tag{2}$$

where y is the unemployment risk indicator or the expected tenure in a given occupation, faced by worker i of cohort g (the cohort is defined by the treatment year) in year $t \in \{2010, ..., 2022\}$, D is a dummy equal to one for individuals who experienced a collective dismissal and zero for individuals who have yet to experience one, e is event time (with $e = 0$ corresponding to the year of treatment, i.e., the year in which workers report their post-displacement employment as their main job[26]), X is a vector of pre-treatment ($e = -1$) time-invariant covariates (including full-time and temporary employment status, macro-region of work, age, gender and education).[27] Standard errors are clustered at the worker level, and the chosen estimation method is the doubly robust difference-in-differences estimator based on stabilized inverse probability weighting and ordinary least squares

26 In the job-displacement literature, the treatment year is conventionally defined as the calendar year in which a worker is displaced. Earnings losses are typically already observable in that year, although they often deepen further in the subsequent period. I depart from this convention in order to identify occupational switches more cleanly. In the analysis sample, defining the treatment year as the calendar year in which job loss occurs would imply that, for roughly 75 percent of displaced workers, the main occupation recorded in the displacement year would still be the occupation from which they were displaced. This would mechanically attenuate treatment effects in the first post-treatment year. Nevertheless, I report results based on this conventional treatment-timing definition in Figure B4 of Appendix B.

27 Notice that for the analysis that exploits never-treated units as controls, I fixed covariates to the last year before any unit in the sample had been treated, which is 2012.

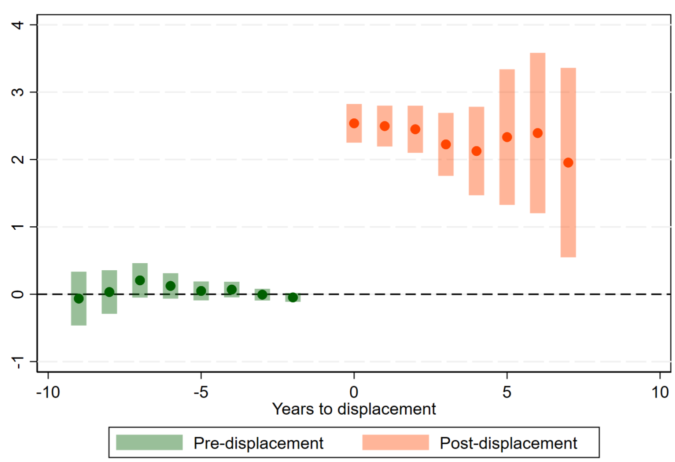
as in Sant'Anna and Zhao (2020). The parameter of interest is β, which can be interpreted as the effect of displacement on the job security (in terms of either unemployment risk or expected tenure) of the subsequent occupations found by dismissed workers. As discussed in the Data section, these effects should be interpreted with the caveat that they are estimated only for workers who report at least one employment spell in each year over the entire period considered, thereby excluding the long-term unemployed. Nevertheless, because the analysis excludes the workers who are the most adversely affected by displacement, the resulting estimates are, if anything, conservative.

Finally, to benchmark the results, I performed the estimation using a stacking-by-event methodology as in Deshpande and Li (2019) or Cengiz et al. (2019), though with an unconditional specification alongside cohort-worker and cohort-year fixed effects.

5 Results

Figure 2 presents the results obtained from estimating equation (2) for the first of the two outcome variables, the unemployment risk. The chart displays the expected change in the unemployment risk linked to the occupations held by treated units compared to the variation in the control group before and after the treatment. In other words, post-treatment estimates reflect the average treatment effect of displacement on the expected unemployment risk associated with the occupations that workers find after losing their jobs.

Figure 2: Unemployment risk



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

All pre-treatment coefficients are close to zero and not significant, which suggests that the parallel trends assumption holds. Post-treatment estimates are, instead, all positive, significant, and relatively stable over time, ranging between 2.56 pp in the first post-treatment period and 2.08 pp in the last period. Note that the chart omits the value corresponding to $e = 8$, which is also excluded from all regression results presented throughout the rest of the analysis. The reason for not showing this estimate is that its value is obtained by exploiting only one treated cohort (that of individuals treated in 2013) and one not-yet-treated cohort as a control group (that of those treated in 2022). However, as reported in Table A7 in Appendix A, the 2013 cohort accounts for 1165 individuals, while the cohort of 2022 counts only 236 individuals. This implies that there is about one control unit for every five treated ones. Given the limited size of the last-treated cohort and the relative imprecision of this estimate, I preferred to omit it from the presented results. An analogous motivation applies to two pre-treatment coefficients that have also been discarded from the chart, namely those corresponding to the periods $e = -10$ and $e = -11$.

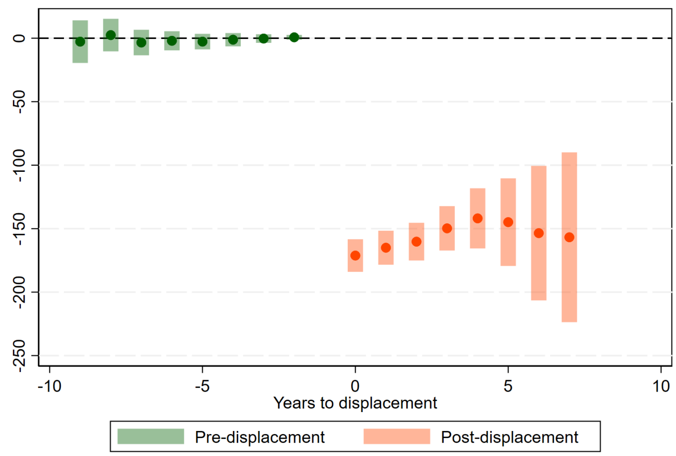
Estimates obtained using never-treated individuals as controls (shown in Figure B5 of Appendix B) are similar to those retrieved with not-yet-treated units, although coefficients are slightly smaller in magnitude. Additionally, while the average of the pre-treatment estimates is close to zero, as in the previous case (0.125), they are jointly significant in the never-treated specification, even though this result is mainly driven by the coefficients that are farther away from the treatment period. Taking stock of these findings, while using never-treated units as controls yields post-treatment estimates that are comparable to those obtained with not-yet-treated units, an examination of pre-trends suggests that, as argued in the previous section, the latter type of control units is more appropriate for the empirical setting under consideration.

Figure 3 reports the corresponding estimates for expected tenure, which can be interpreted analogously to those in Figure 2. Again, all pre-treatment coefficients are not significant and close to zero, suggesting that the parallel trends assumption holds. Post-treatment estimates are, instead, negative, significant, and relatively steady over time, ranging from a maximum of 171 days of reduced expected tenure in the first post-treatment year to a minimum of 142 days in the fifth.

Estimates obtained using never-treated units as a control group are shown in Figure B6 in Appendix B. Similarly to what I observed for the unemployment risk indicator, the magnitude of these coefficients is generally smaller than that of estimates shown in Figure 3, and pre-treatment coefficients are jointly significant but relatively close to zero, showing the same sign as post-treatment coefficients. Combined with the results for the unemployment risk indicator, I interpret these findings as providing further indication that not-yet-treated units constitute more suitable counterfactuals to extrapolate potential outcomes in this empirical framework.

Moreover, the analysis has been replicated using a stacking-by-event approach. As mentioned in the previous section, this specification does not feature covariates. Yet, the findings are in line with the baseline results and are reported in Table A8 of Appendix A. These estimates can also be compared with those derived using unconditional specifications through the CS methodology, which are plotted in Figures B7 and B8 of Appendix B. The main takeaway from these latter results is that adding covariates to retrieve credible conditional parallel trends seems to matter only for the expected tenure indicator that uses never-treated units as controls, which further reinforces the case for the not-yet-treated specification.

Figure 3: Expected tenure



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in expected tenure associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Overall, all specifications consistently suggest that displacement causes a substantial decline in occupational job security. To give a sense of the magnitude of these effects, the increase in unemployment risk corresponds to a 12.8% rise relative to the pre-displacement average, while the reduction in expected tenure corresponds to an approximately 11.7% drop. Both indicators thus point to comparable reductions in occupational job security, corroborating one another.

On a final note, before presenting heterogeneous results for different subgroups of workers, two other dimensions of heterogeneity are worth mentioning. First, cohort-specific estimates reveal that the impact of dismissal varies substantially across cohorts. The 2019 cohort appears to be the most adversely affected, with post-treatment averages of 4.22 pp for the unemployment risk indicator and a reduction of 254 days in expected tenure. In contrast, the 2017 and 2016 cohorts are the least affected, with post-treatment averages of 1.12 pp and 1.83 pp for the unemployment risk indicator and reductions of 119 and 116 days in expected tenure, respectively. A second dimension of heterogeneity stems from partitioning the sample into workers who transitioned to their next main occupation within six months of dismissal and those who took longer to do so. Notice that, as outlined in the data section, the dataset is constructed to retain only a single dominant employment record per worker for each year. Consequently, it is not easy to precisely calculate the number of unemployment days between the date of the collective dismissal and the employment that immediately follows. Therefore, for the sake of simplicity, here I consider the number of days between two consecutive main employment spells as a proxy for the time it takes a displaced worker to secure a new, relatively stable position. Furthermore, since the partitioning of the sample relies on post-treatment outcomes, which may introduce non-trivial selection of workers, the resulting evidence should be viewed as purely descriptive. Bearing these caveats in mind, given that human capital depreciates over time and employers often attach a stigma to long-term unemployment, it is reasonable to expect that individuals who take longer to find a suitable job also experience worse outcomes. Indeed, workers who find their next main job within six months experience substantially weaker effects than those who take longer, even though confidence bands overlap in later periods, which are more imprecisely estimated. These results are presented in Figure B9 in Appendix B.

5.1 Heterogeneity of results

The findings in the previous section document a significant and persistent impact of displacement on occupational job security. A natural follow-up question is whether these adverse effects vary across different subgroups of workers. I therefore examine the impact of job loss by gender, educational attainment, and region of employment to assess the presence of heterogeneous treatment effects.[28]

First, I ask whether there is a gender differential in the impact of job loss on occupational job security. Based on the evidence in the literature, the answer to this question is not straightforward. In fact, on the one hand, some studies have shown that women tend to sort into jobs with lower wage premiums (Card et al., 2016) but also a reduced risk of dismissal (Wilkins and Wooden, 2013; see also Table A4). On the other hand, some recent analyses have revealed that job loss leads to more substantial and persistent reductions in earnings for women (Illing et al., 2024). Taken together, these findings point to two hypotheses that could explain the observed gender gap in the earnings cost of job loss. First, compared with their male counterparts, women may end up accepting lower-paid jobs after dismissal, with limited wage-growth potential (e.g., part-time jobs). This hypothesis would be consistent with a scenario in which the observed gender gap in earnings following job loss originates primarily from wages or hours worked. Second, displacement may erode women's relative advantage over men in terms of job security. In this latter case, the gender gap in earnings due to displacement would come from a larger decline in job security, reflecting effects on prospective employment stability. Naturally, while both could be at work simultaneously, the results presented next test only the latter mechanism.

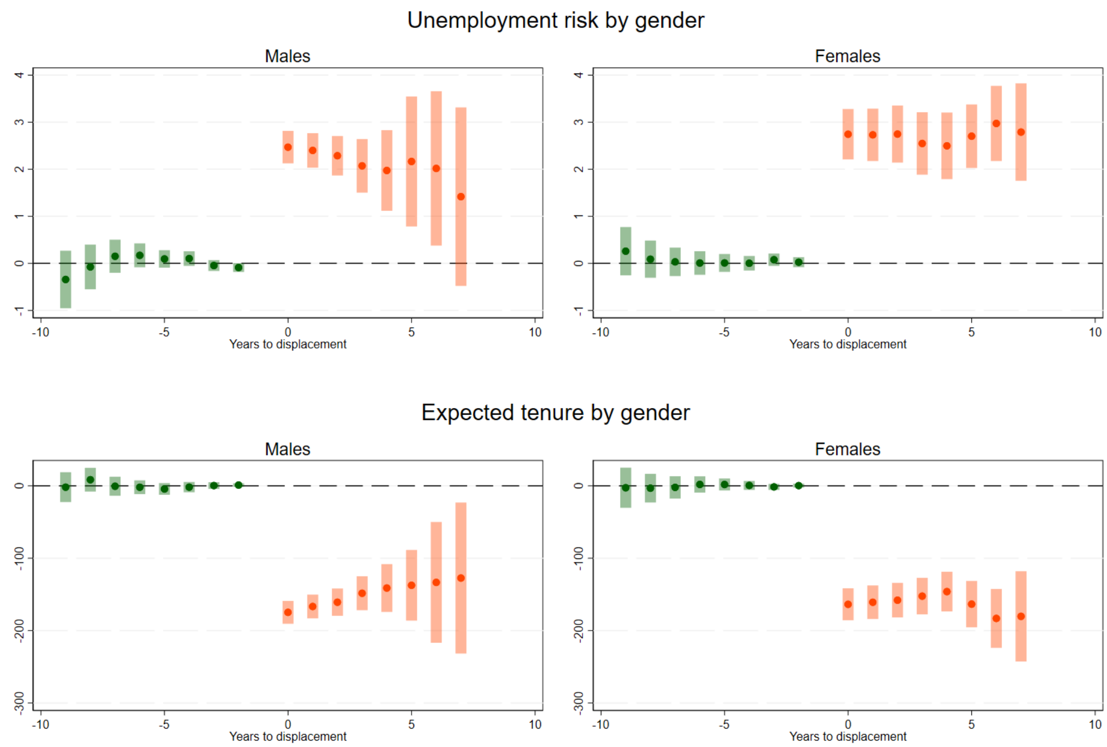
Figure 4 presents estimates[29] for males on the left and females on the right. The chart in the top-right part of the panel shows that females experience an average increase in unemployment risk of approximately 2.72 pp, while males (top-left corner of the chart) face a comparatively smaller average increase of about 2.1 pp. Declines in expected tenure align with these results, amounting to about 163

28 For the regressions based on the region of work, workers were classified according to the macro-region (namely, South, Center, and North of Italy) where they were employed in the year prior to the treatment, since the region of work, unlike gender and education, may change over time.

29 These estimates, like those shown in Figures 5 and 6, are derived without including the educational level as a covariate. This decision was made because, when splitting the sample to estimate the model on separate subgroups, the sample size of certain cohorts was not sufficiently large to accommodate all the covariates in the original model. Nevertheless, excluding education from the model does not seem to alter substantially the coefficient of interest.

days for females and 149 days for males. These averages, however, mask rich dynamics in post-treatment effects. In the immediate aftermath of displacement, the drop in occupational job security seems to affect men and women to a similar extent. Over time, however, these negative effects become moderately larger for women, while the opposite pattern holds for men.

Figure 4: Heterogeneous effects of displacement on job security by gender



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Estimates derived using never-treated units (shown in Figure B10 in Appendix B) quantitatively resemble those presented in Figure 4. Both specifications convey a very similar message: while the negative impact of dismissal on both outcomes is initially almost equivalent in magnitude for men and women, these effects gradually attenuate over time for males but slightly intensify for females.[30]

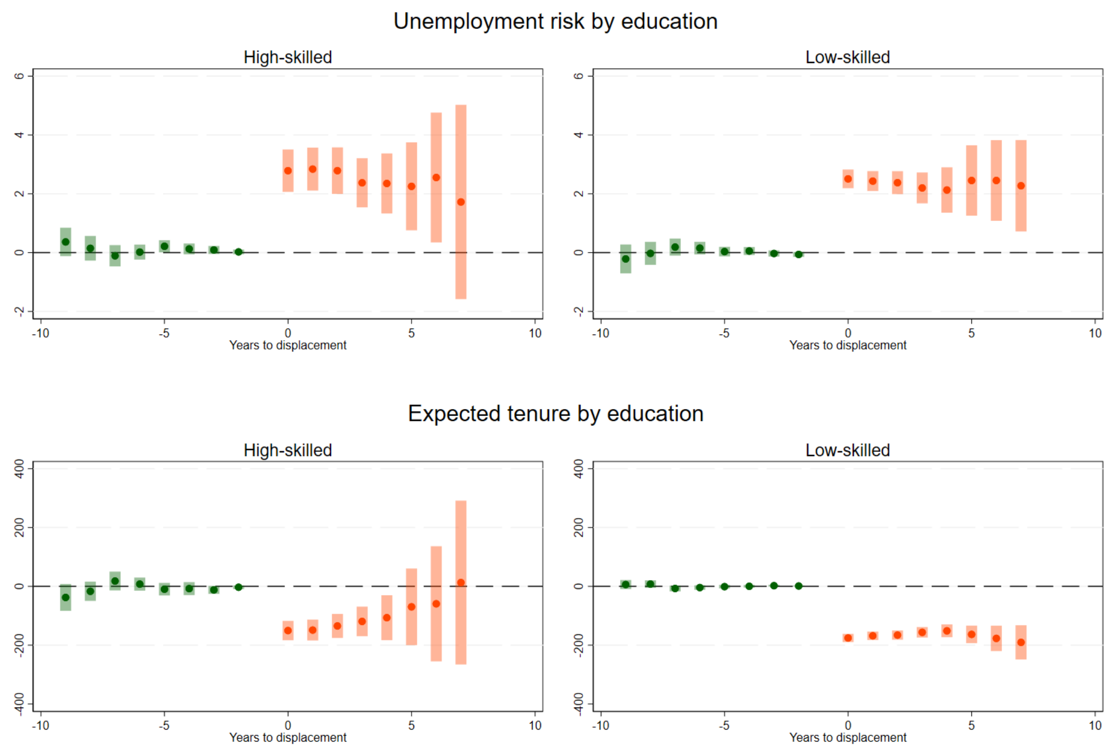
It is important to emphasize, nonetheless, that these considerations apply only when point estimates are considered in isolation. In fact, confidence intervals are too wide to establish any statistically significant differences between men and women. Therefore, these results do not provide sufficient evidence to support the hypothesis that women, after dismissal, lose their edge over men

30 More precisely, in the never-treated specification, the expected tenure indicator suggests that women, like men, modestly recover from the negative effects of dismissal over time. However, their recovery is slower, discontinuous, and less pronounced. Altogether, the patterns suggested by this indicator are less clear than in the not-yet-treated specification.

in terms of occupational job security, even though the point estimates do suggest that this might be happening to some moderate degree.

Next, I examine the presence of heterogeneous effects by educational level. To do so, I categorized individuals as high-skilled if they obtained at least a college degree and low-skilled otherwise. The results are displayed in Figure 5.

Figure 5: Heterogeneous effects of displacement on job security by educational level



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Differences between higher- and lower-educated individuals seem to emerge only in later periods and only for the expected tenure indicator. On average, the expected tenure of lower-educated individuals decreases by approximately 168 days after displacement, whereas the decline is less pronounced for higher-educated individuals, at around 97 days. This substantial difference in post-treatment averages primarily reflects the recovery observed among higher-educated individuals, who appear to continue recovering from the negative effects of displacement over time. In contrast, no notable differences can be observed between these two subgroups for the other indicator.[31] As for the previously analyzed case, confidence intervals are too wide to assert the presence of statistically sig-

31 Although the last post-treatment estimate for highly educated individuals is not significant, this is likely attributable to limited statistical power.

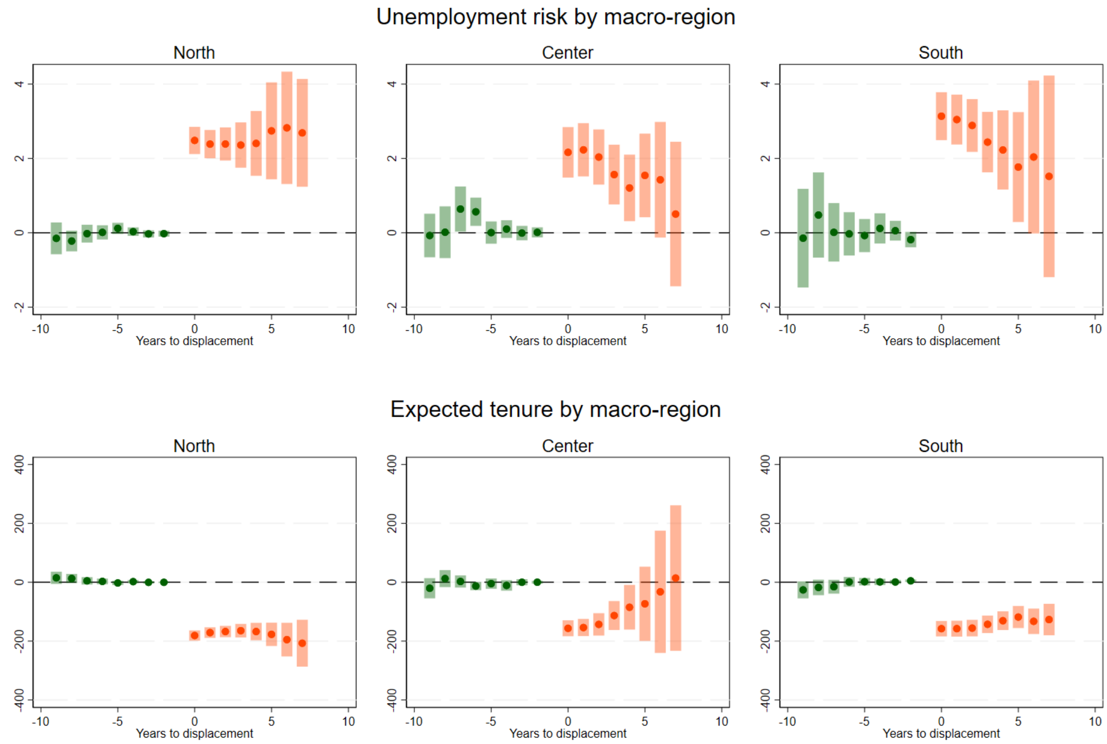
nificant differences between these two categories. Furthermore, Figure B11 in Appendix B, reporting results obtained with never-treated units, does not clarify whether higher-educated individuals recover faster than lower-educated ones from the negative effects of displacement. Weighing all these elements, these findings provide only a tentative indication that high-skilled individuals may be able to reverse the negative effects of displacement over time. Thus, I conclude that no meaningful differences can be established between educational groups.

Lastly, Figure 6 displays the results from separate regressions by macro-region of work. Again, it is not possible to detect statistically significant differences among these three subgroups because of wide confidence intervals. Nevertheless, judging solely from the point estimates, individuals employed in the North of Italy at the time of dismissal seem to suffer harsher and more persistent consequences from job loss, reporting a rise in unemployment risk of about 2.53 pp and a reduction in expected tenure of about 179 days. An inspection of pre-treatment averages (shown in Table A6), however, reveals that individuals working in the northern regions of Italy are generally employed in jobs with a substantially lower risk of dismissal (about three percentage points less than in the Center and five percentage points less than in the South) and a considerably higher expected tenure (about 14% higher than in the Center, and 24% higher than in the South). These pre-treatment differences suggest that individuals working in the North have more to lose from dismissal than their counterparts in other regions, potentially explaining, at least partially, the patterns observed in Figure 6. A simple back-of-the-envelope calculation (obtained by adding the average treatment effect of displacement to the pre-treatment averages, or subtracting it in the case of the expected tenure indicator) shows that, although workers in the North experience harsher effects from displacement, the average occupation they find afterward still carries substantially higher job security than the average occupation found by workers in the Center and the southern regions of Italy. Moreover, given that unemployment rates in the North of Italy are historically lower than in the rest of the country, these results are consistent with the ”stigma effects” theorized in the literature.[32]

Finally, it is important to highlight that the narrative conveyed by these estimates is not consistent with the one suggested by the estimates obtained using never-treated units as the control group, as shown in Figure B12 in Appendix B. In this latter case, the evidence regarding heterogeneous treatment effects across different macro-areas in Italy appears more mixed. Since confidence intervals do not allow me to make definitive statements about the regional heterogeneity of these effects, and given the inconsistency between the results of the two types of specifications, these findings should be viewed with caution.

32 Omori (1997) shows that workers who experience nonemployment when only a few workers are nonemployed are more severely stigmatized.

Figure 6: Heterogeneous effects of displacement on job security by macro-region



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

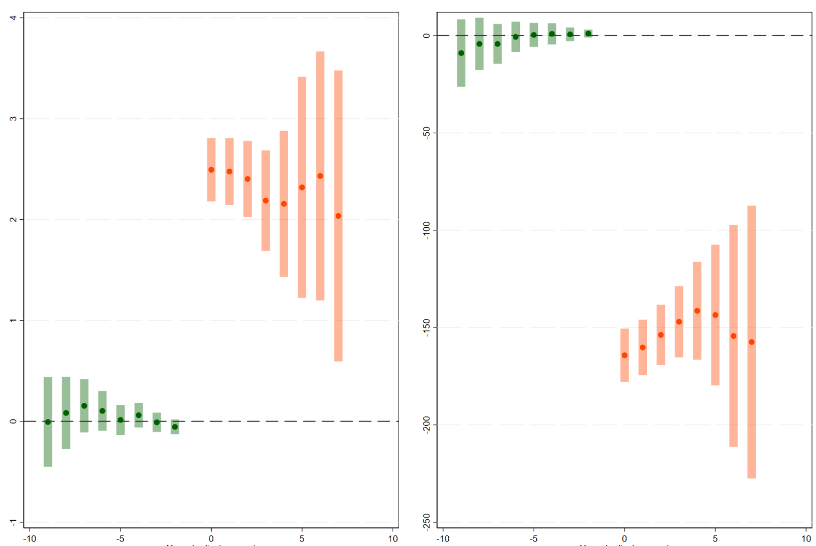
6 Robustness and placebo

As a final step, I test the soundness of the presented results through a series of robustness checks and sensitivity analyses addressing the following concerns: i) some sectors and occupations are less represented in the sample, potentially affecting the accuracy of the constructed indicators of occupational job security; ii) the indicators were calculated by including episodes of collective dismissals as events of involuntary job loss, which may raise endogeneity concerns; iii) collective dismissals may generate spillover effects in local labor markets, potentially amplifying the estimated effects of displacement; iv) a small number of outlier observations may be driving the estimated results; v) since the main estimates are obtained from a sample of workers aged between 30 and 54, expanding the sample to include both younger and older workers could yield substantially different results. Finally, as a placebo test, I arbitrarily reassigned the treatment to three years before its actual occurrence to verify that a significant change in the outcome variable occurred around the time of treatment. I will present these exercises in the following subsections.

6.1 Robustness checks

The first check addresses concerns that may arise from the fact that some sectors and occupations are underrepresented in the sample, potentially making the computation of job security indicators imprecise. To mitigate these concerns, I excluded from the analysis sectors and occupations with fewer than one hundred available observations per year in the original sample, retaining only those for which both outcome measures could be computed with higher precision. Consequently, individuals employed in these excluded occupations at any point during the sample period were also discarded. After these operations, the sample size decreased to 94,653 observations. The estimates resulting from this refined sample are plotted in Figure 7, closely resembling those from the main specification and differing only slightly in magnitude.

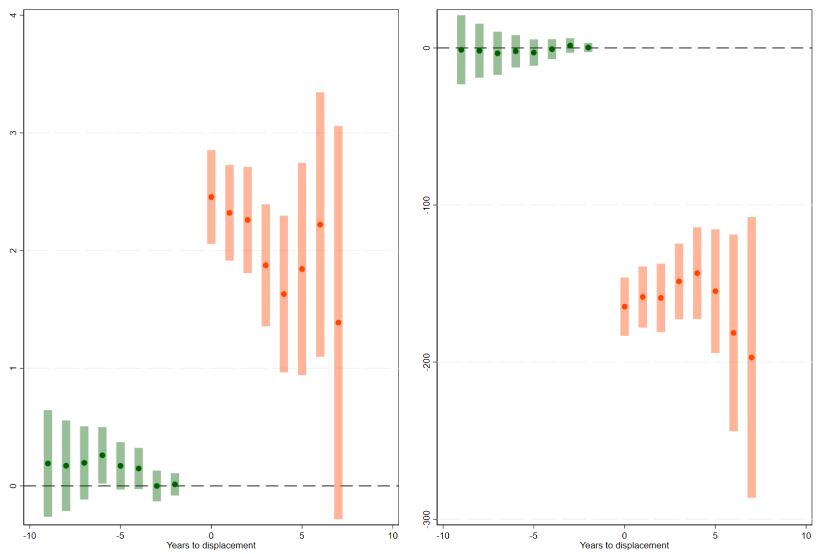
Figure 7: Robustness: excluding occupations and sectors with less than one hundred observations per year in the original sample



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

A second source of concern might stem from the inclusion of collective dismissals in the construction of the proposed occupational job security indicators, as this may raise endogeneity concerns. Two main considerations motivated this choice. First, excluding these events would have misrepresented the inherent security of the affected occupations. Second, these are the occupations workers leave behind when they get dismissed. Since incorporating collective dismissals into the construction of the job security measures can only make these occupations appear less secure (regardless of workers' subsequent occupations), this approach is likely only to induce a downward bias in the coefficients of interest. Nevertheless, to address this concern, I implemented the following procedure. First, I randomly split the original sample into two groups and re-computed the job security indicators using only one of them. Then, I re-estimated the results using the subset of treated individuals from the group that was not used for the computation of the indicators. This subset consists of 53,885 observations, approximately half the sample used for the main results. The outcome of this exercise is shown in Figure 8. Owing to the shrinkage in the sample size, these estimates are less precise than the main results. Still, post-treatment coefficients are not significantly different from those of the main specification, though the unemployment risk indicator exhibits a sharper recovery over time, while the expected tenure indicator appears to move in the opposite direction. Overall, these findings confirm that the main results remain robust even when the job security indicators and estimates are derived from two disjoint sets of observations.

Figure 8: Robustness: split sample



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Lastly, a further concern relates to the use of large-scale layoffs as a source of exogenous variation, which could be associated with significant spillover effects in local labor markets, ultimately amplifying the measured negative effects of displacement. Indeed, widespread layoffs in a particular region might have significant repercussions for local labor demand, making it more difficult for unemployed workers to find new, stable jobs. As outlined in previous sections, collective dismissals are usually smaller in scale than conventionally defined mass layoffs, which already partially alleviates these concerns. Nevertheless, to directly address them, I add regional unemployment rates to the main specification to control for these potential spillovers and re-estimate the model.[33] As the CS estimator does not work well with time-varying covariates, I add this control to the regression specification estimated using the methodology introduced by Cengiz et al. (2019). The resulting estimates are remarkably close to those generated by the main specification, and are reported in Table A11 in Appendix A.

6.2 Sensitivity of results

One may also worry that outlier observations might be driving the results. To rule out this possibility, I excluded occupations that fall in the top or bottom 1% of the distributions for either the unemployment risk or the expected tenure indicator. As before, this also excludes all individuals employed in these occupations at any point during the sample period. This operation removes about 3% of the total sample, reducing the sample size to about 104 thousand observations. The estimates, shown in Figure B13 in Appendix B, are similar to those retrieved from the main specification and even slightly higher in magnitude.

In a second sensitivity check, I expanded the sample to include both senior and younger workers, encompassing all individuals aged 20 to 64 years. This operation increased the sample size to approximately 195 thousand observations. The rationale for focusing only on prime-age workers in the main analysis was that both young and senior workers may respond to displacement very differently from prime-age workers. Young workers, whose careers are just beginning, may suffer more from unemployment because it prevents them from building up the human capital that employers demand. On the other hand, senior workers, being closer to retirement, may face significant challenges in finding new employment, as late-career vacancies may be limited or require up-to-date skills that they lack. The estimates of this exercise are displayed in Figure B14 in Appendix B. Overall, these estimates corroborate the findings presented throughout the study. The unemployment risk indicator, however, exhibits a distinct pattern, with treatment effects intensifying in later periods.

6.3 Placebo test

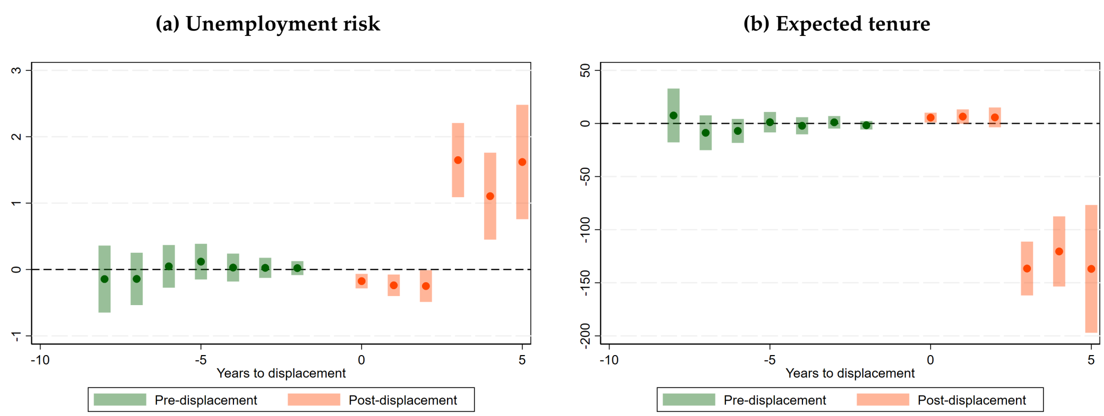
Finally, as a placebo test, I reassigned treatment to three years before its actual occurrence. The specific number of years was chosen arbitrarily and could have been set to any number of years preceding the actual treatment.[34] Figure 9 shows the results from this exercise. The chart on the left depicts a very small (about 0.22 pp on average) yet statistically significant negative effect for the three periods following the placebo treatment. However, while these estimates are all significant, they are also of the opposite sign relative to those for the true treatment. If anything, this may suggest that, before treatment, individuals were on a positive trajectory, gaining occupational job security over

33 This covariate was not included in the main specification because it might represent a "bad control," as it is likely to be endogenous to the treatment variable and potentially bias the estimates. Nevertheless, for the specific purpose of assessing the influence of potential spillovers generated by large-scale layoffs on the main results, concerns about causality can be set aside temporarily.

34 I replicated this placebo test by retiming treatment to two years before the actual treatment. The results of this exercise are shown in Figure B15 in Appendix B.

time. Moreover, these small negative effects vanish when the placebo treatment is moved, for example, to two years before the real treatment (see Figure B15 in Appendix B). Similar reasoning applies to the expected tenure indicator. In both cases, the real effect emerges only three years after the placebo treatment, aligning with the timing of the actual displacement year. These results reinforce the hypothesis that the findings in the previous section do capture a fundamental change occurring in conjunction with the year of treatment.

Figure 9: Placebo displacement



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which ”treated” individuals are employed each year. Estimates are obtained using ”placebo” not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

7 Conclusions

This paper provides new insights into the costs of job loss by analyzing how displacement affects the job security of subsequent occupations. For this purpose, I construct two measures of occupational job security and empirically estimate the causal impact of displacement on these outcomes. While previous research has extensively investigated the effects of displacement on workers' earnings, wages, and future unemployment, this study narrows the focus to the characteristics of the jobs into which displaced workers sort after job loss.

Focusing on variation in occupational job security is particularly relevant for two reasons. First, it quantifies the extent of the downward move along the occupational job-security ladder, isolating, as much as possible, demand-side factors in the post-displacement labor market from supply-side ones. Second, it highlights occupational sorting as a potential channel through which displaced workers face heterogeneous exposure to recurrent unemployment. Taken together, the findings hint at underlying demand-side deficiencies in the post-displacement labor market that may contribute to the persistent effects of job loss documented in the literature.

This study finds a significant decline in occupational job security. On average across post-treatment periods, post-displacement occupations are associated with an increase in unemployment risk of approximately 2.36 percentage points and a decrease in expected tenure of about 155 days. The dynamics of the estimated treatment effects indicate a gradual, albeit modest, recovery over time. While point estimates suggest some heterogeneity by gender, educational level, and region of employment, the results do not provide conclusive evidence of differential effects.

In light of these findings, some policy implications may be drawn, although they should be viewed with caution given that the paper remains agnostic about the mechanisms underlying the observed sorting of displaced workers into new occupations. To mitigate the costs of job loss and facilitate workers' reintegration into the labor market, vocational training is a policy response that is commonly advocated and implemented. While vocational training may alleviate these costs, it may not fully offset the loss of occupation-specific human capital that occurs when displaced workers change occupations. Moreover, finding a new occupation plausibly requires workers not only to acquire new skills but also to exert considerable effort in signaling unobservable qualities to potential employers. Combined with poor outside options, such frictions could lead displaced workers to accept occupations that entail lower job security, consistent with the persistently weaker employment prospects documented in this study.

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A Appendix

Table A1: Distribution of individuals in the treatment group across genders

Gender	N	Percent
Male	5562	67.37
Female	2694	32.63

Table A2: Distribution of individuals in the treatment group across educational levels

Level of education	N	Percent
None or elementary school	172	2.09
Middle school	2353	28.50
High school	4309	52.19
University degree or more	1422	17.22

Table A3: Distribution of individuals in the treatment group across macro-regions of Italy

Macro-region	N	Percent
North	4822	58.41
Center	1681	20.36
South and Islands	1753	21.23

Table A4: Mean value of the outcome variables in the treatment group across genders in pre-treatment years

Gender	Unemployment risk	Expected tenure
Male	18.55	1331.78
	(11.07)	(611.65)
Female	18.19	1327.89
	(11.42)	(576.94)

Notes: Standard errors in parentheses.

Table A5: Mean value of the outcome variables in the treatment group across educational levels in pre-treatment years

Level of education	Unemployment risk	Expected tenure
None or elementary school	24.54	972.61
	(14.40)	(520.21)
Middle school	20.90	1179.17
	(11.60)	(603.61)
High school	17.58	1381.29
	(10.62)	(583.74)
University degree or more	16.34	1459.00
	(10.90)	(589.15)

Notes: Standard errors in parentheses.

Table A6: Mean value of the outcome variables in the treatment group across macro-regions of Italy in pre-treatment years

Macro-region	Unemployment risk	Expected tenure
North	16.73	1422.99
	(9.11)	(588.92)
Center	19.90	1249.76
	(12.54)	(603.83)
South and Islands	21.76	1150.13
	(13.82)	(575.30)

Notes: Standard errors in parentheses.

Table A7: Distribution of observations across cohorts of treated individuals

Year of treatment	Number of treated ids	Percentage
2013	1165	14.11
2014	1459	17.67
2015	1742	21.10
2016	1218	14.75
2017	915	11.08
2018	539	6.53
2019	411	4.98
2020	367	4.45
2021	204	2.47
2022	236	2.86
Total	8256	100.00

Table A8: Results: Unconditional regression, stacking à la Cengiz et al. (2019) - control group: not-yet-treated units

	Unemployment risk	Expected tenure
Tm9	-0.195	7.948
	(0.20)	(8.24)
Tm8	-0.115	12.799*
	(0.17)	(7.09)
Tm7	0.237	-1.832
	(0.15)	(5.85)
Tm6	0.250**	-5.975
	(0.11)	(4.66)
Tm5	0.058	-1.573
	(0.09)	(3.64)
Tm4	0.010	0.644
	(0.07)	(2.68)
Tm3	0.010	-0.014
	(0.04)	(1.40)
Tm2	-0.024	0.223
	(0.03)	(0.87)
Tp0	2.545***	-169.938***
	(0.15)	(6.52)
Tp1	2.419***	-161.449***
	(0.15)	(6.72)
Tp2	2.282***	-155.371***
	(0.16)	(6.95)
Tp3	2.145***	-148.856***
	(0.17)	(7.28)
Tp4	2.140***	-144.344***
	(0.19)	(7.88)
Tp5	2.243***	-143.003***
	(0.23)	(8.91)
Tp6	2.304***	-140.290***
	(0.30)	(10.41)
Tp7	2.231***	-136.780***
	(0.36)	(12.64)
Observations	279,924	279,924

Notes: The table above displays estimates obtained via a stacking-by-event specification à la Cengiz et al. (2019). Standard errors are reported in parentheses below each coefficient. Stars indicate p-values, with: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

Table A9: Results: conditional regression à la Callaway and Sant'Anna (2021) - control group: not-yet-treated units

	Unemployment risk	Expected tenure
Tm9	-0.083	-2.709
	(0.208)	(8.562)
Tm8	0.011	2.472
	(0.167)	(6.543)
Tm7	0.146	-3.406
	(0.128)	(5.133)
Tm6	0.139	-2.039
	(0.094)	(3.835)
Tm5	0.071	-2.687
	(0.071)	(3.124)
Tm4	0.071	-1.188
	(0.059)	(2.668)
Tm3	-0.004	-0.264
	(0.045)	(1.725)
Tm2	-0.047	0.713
	(0.034)	(0.961)
Tp0	2.559***	-171.188***
	(0.148)	(6.553)
Tp1	2.510***	-165.015***
	(0.157)	(6.824)
Tp2	2.458***	-160.230***
	(0.181)	(7.575)
Tp3	2.254***	-149.741***
	(0.243)	(8.917)
Tp4	2.171***	-141.890***
	(0.345)	(12.083)
Tp5	2.397***	-144.877***
	(0.536)	(17.600)
Tp6	2.431***	-153.535***
	(0.620)	(27.031)
Tp7	2.082***	-156.790***
	(0.743)	(34.121)
Observations	107,328	107,328

Notes: The table reports the point estimates displayed in Figures 2 and 3 relative to the specifications run with not-yet-treated individuals as control units. Standard errors are reported in parentheses below each coefficient. Stars indicate p-values, with: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

Table A10: Results: conditional regression à la Callaway and Sant'Anna (2021) - control group: never-treated units

	Unemployment risk	Expected tenure
Tm9	0.461*	-23.089**
	(0.252)	(9.042)
Tm8	0.056	-8.566
	(0.145)	(6.086)
Tm7	0.189	-9.842**
	(0.115)	(5.009)
Tm6	0.135*	-7.633**
	(0.076)	(3.089)
Tm5	0.051	-5.449**
	(0.056)	(2.321)
Tm4	0.067	-3.123*
	(0.049)	(1.719)
Tm3	0.036	-2.384**
	(0.035)	(1.102)
Tm2	-0.002	-1.354**
	(0.026)	(0.663)
Tp0	2.524***	-165.811***
	(0.150)	(6.723)
Tp1	2.329***	-156.970***
	(0.148)	(6.771)
Tp2	2.184***	-152.638***
	(0.146)	(6.748)
Tp3	2.029***	-147.446***
	(0.144)	(6.744)
Tp4	1.847***	-136.975***
	(0.147)	(6.862)
Tp5	1.787***	-130.334***
	(0.152)	(7.163)
Tp6	2.003***	-134.417***
	(0.164)	(7.630)
Tp7	2.073***	-138.913***
	(0.186)	(8.709)
Observations	3,364,361	3,364,361

Notes: The table reports the point estimates displayed in Figures B5 and B6 relative to the specifications run with never-treated individuals as control units. Standard errors are reported in parentheses below each coefficient. Stars indicate p-values, with: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

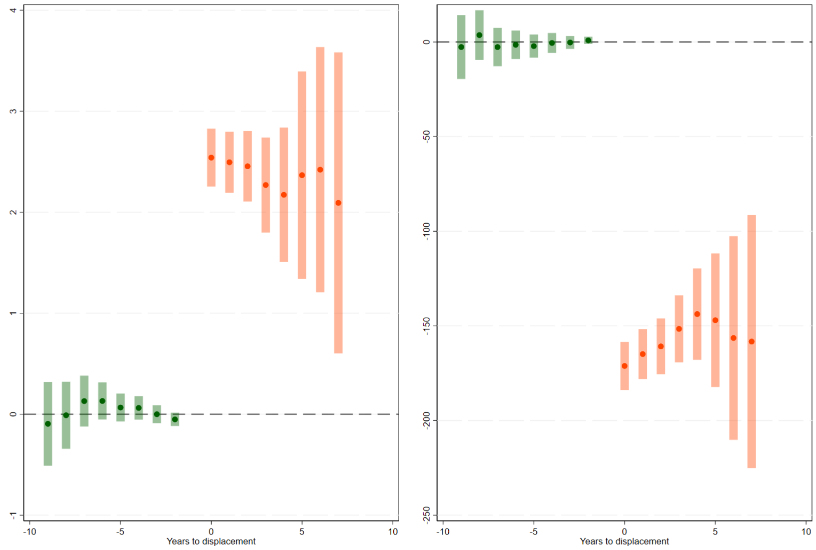
Table A11: Robustness, controlling for regional unemployment rates: Unconditional regression, stacking à la Cengiz et al. (2019) - control group: not-yet-treated units

	Unemployment risk	Expected tenure
Tm9	-0.197	7.983
	(0.20)	(8.24)
Tm8	-0.120	12.887*
	(0.17)	(7.09)
Tm7	0.238	-1.851
	(0.15)	(5.84)
Tm6	0.249**	-5.960
	(0.11)	(4.66)
Tm5	0.056	-1.537
	(0.09)	(3.64)
Tm4	0.008	0.686
	(0.07)	(2.68)
Tm3	0.013	-0.065
	(0.04)	(1.40)
Tm2	-0.020	0.153
	(0.03)	(0.87)
Tp0	2.547***	-169.982***
	(0.15)	(6.52)
Tp1	2.421***	-161.488***
	(0.15)	(6.72)
Tp2	2.284***	-155.414***
	(0.16)	(6.95)
Tp3	2.144***	-148.830***
	(0.17)	(7.28)
Tp4	2.134***	-144.245***
	(0.19)	(7.89)
Tp5	2.235***	-142.853***
	(0.23)	(8.92)
Tp6	2.297***	-140.155***
	(0.30)	(10.42)
Tp7	2.229***	-136.750***
	(0.36)	(12.64)
Observations	279,924	279,924

Notes: The table above displays estimates obtained via a stacking-by-event specification à la Cengiz et al. (2019). Standard errors are reported in parentheses below each coefficient. Stars indicate p-values, with: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

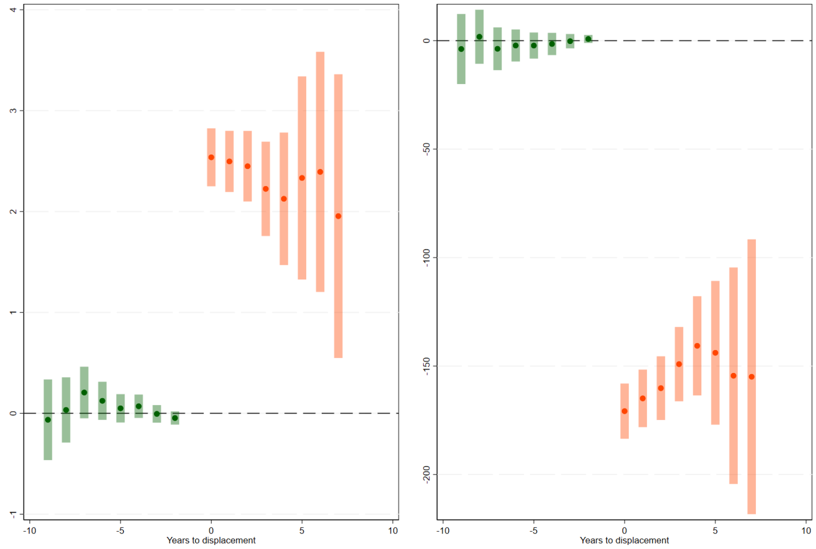
B Appendix

Figure B1: Conditional regression à la Callaway and Sant'Anna (2021) - including workers who lost their jobs more than once due to a collective dismissal procedure



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

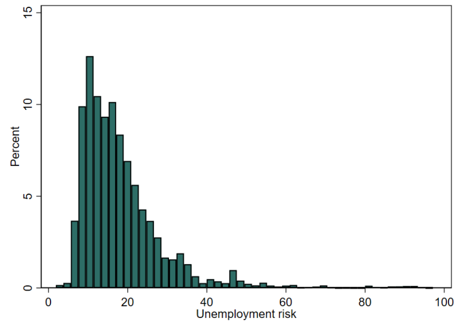
Figure B2: Conditional regression à la Callaway and Sant'Anna (2021) - including workers who found their job before the termination date of their previous one



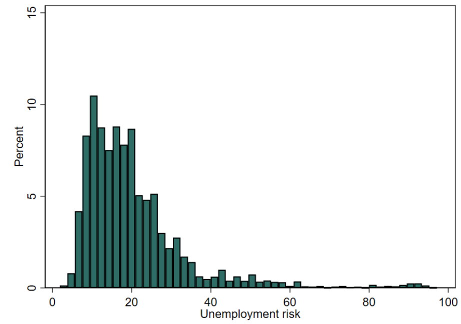
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B3: Pre vs post-treatment distributions of job security indicators

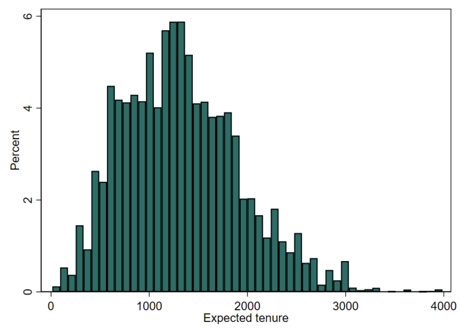
(a) Unemployment risk pre



(b) Unemployment risk post



(c) Expected tenure pre



(d) Expected tenure post

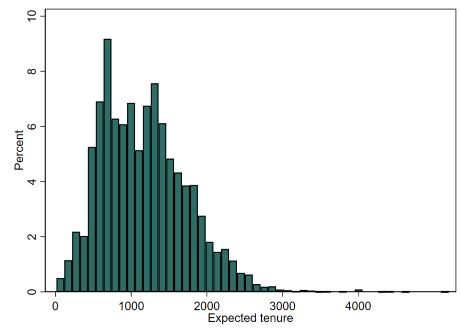
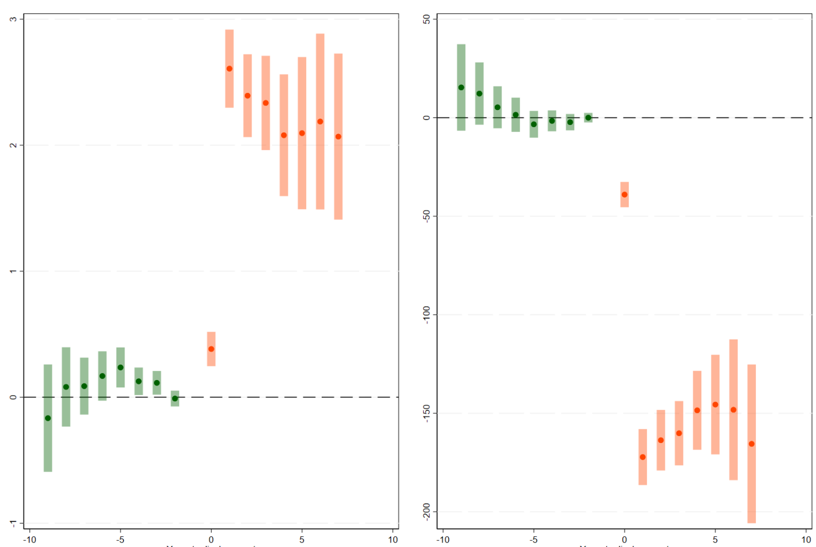
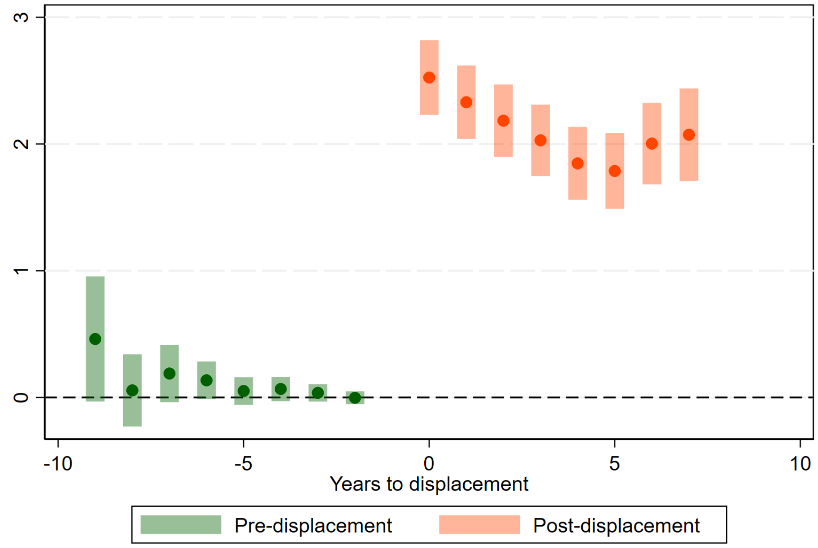


Figure B4: Conditional regression à la Callaway and Sant'Anna (2021) - Conventional treat year



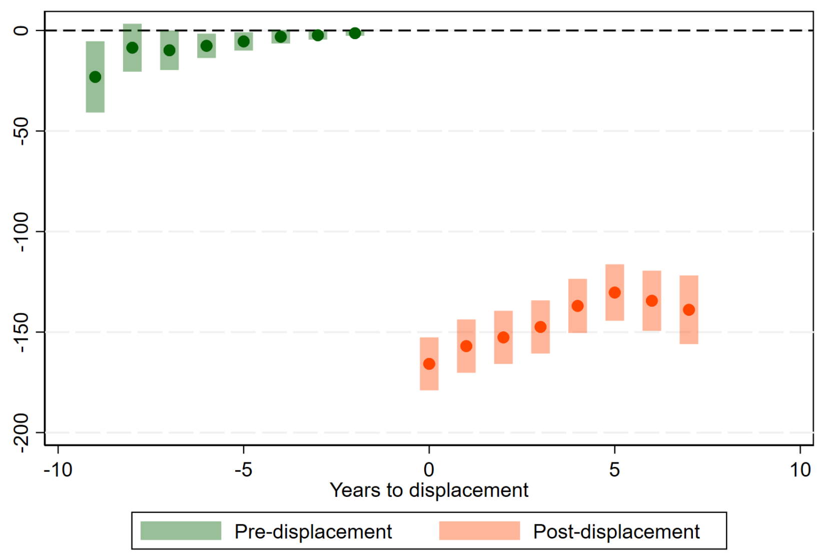
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B5: Unemployment risk - control group: never-treated units



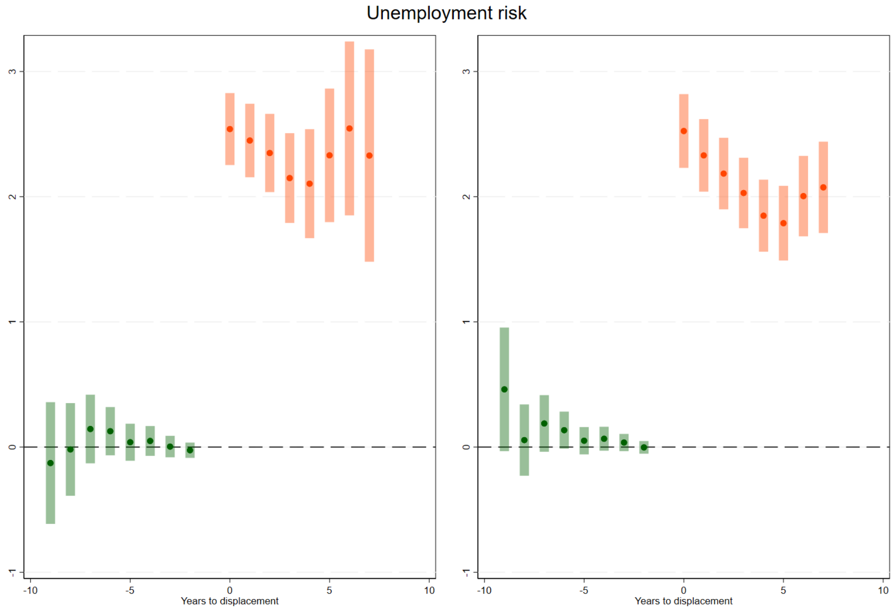
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk associated with the main occupations in which ”treated” individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B6: Expected tenure - control group: never-treated units



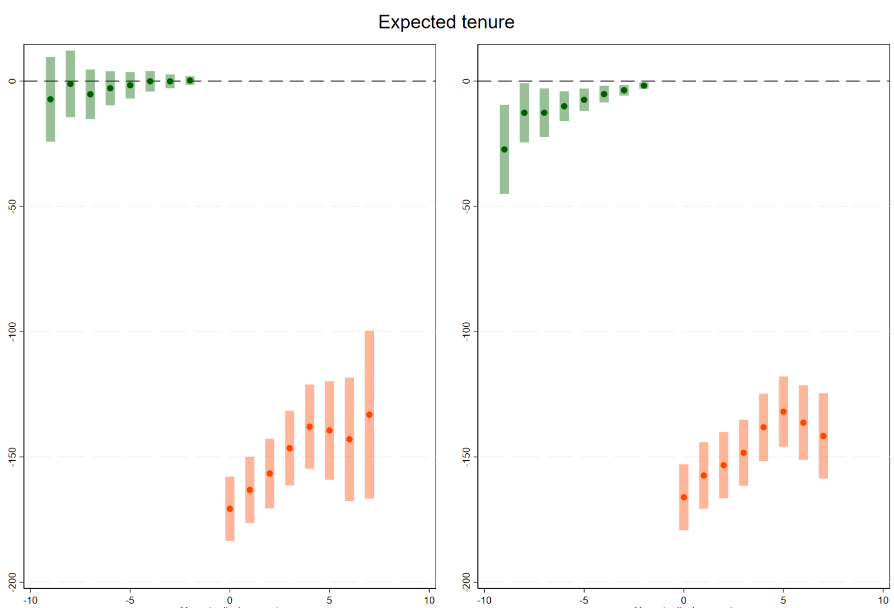
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in expected tenure associated with the main occupations in which ”treated” individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B7: Unconditional regression à la Callaway and Sant'Anna (2021) - Unemployment risk



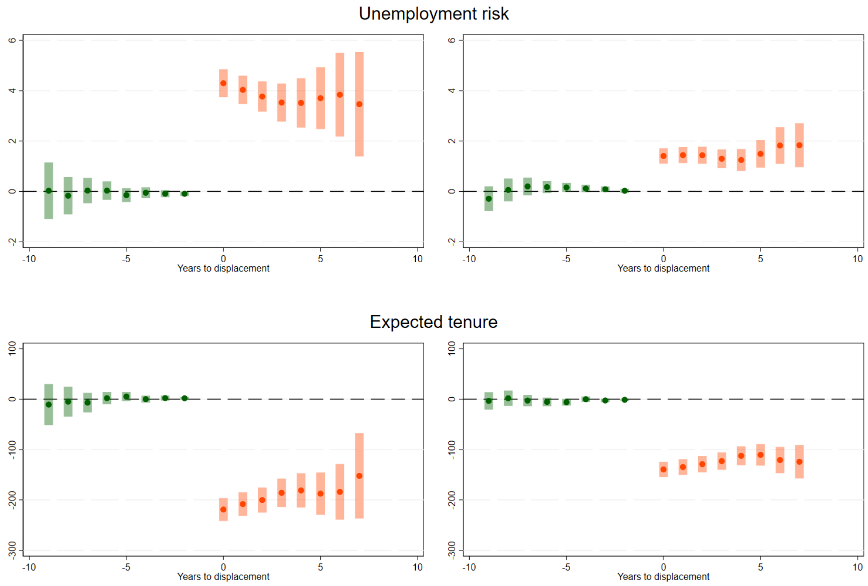
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk associated with the main occupations in which "treated" individuals are employed each year. Estimates on the left panel are obtained using not-yet-treated units as a control group, whereas estimates on the right are obtained using never-treated units. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B8: Unconditional regression à la Callaway and Sant'Anna (2021) - Expected tenure



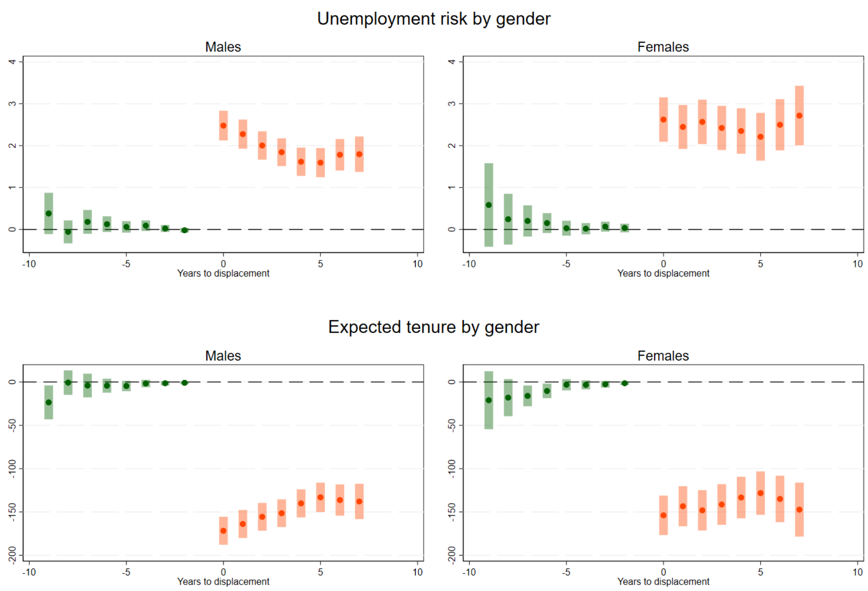
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in expected tenure associated with the main occupations in which "treated" individuals are employed each year. Estimates on the left panel are obtained using not-yet-treated units as a control group, whereas estimates on the right are obtained using never-treated units. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B9: Conditional regression à la Callaway and Sant'Anna (2021) by period length between main occupations



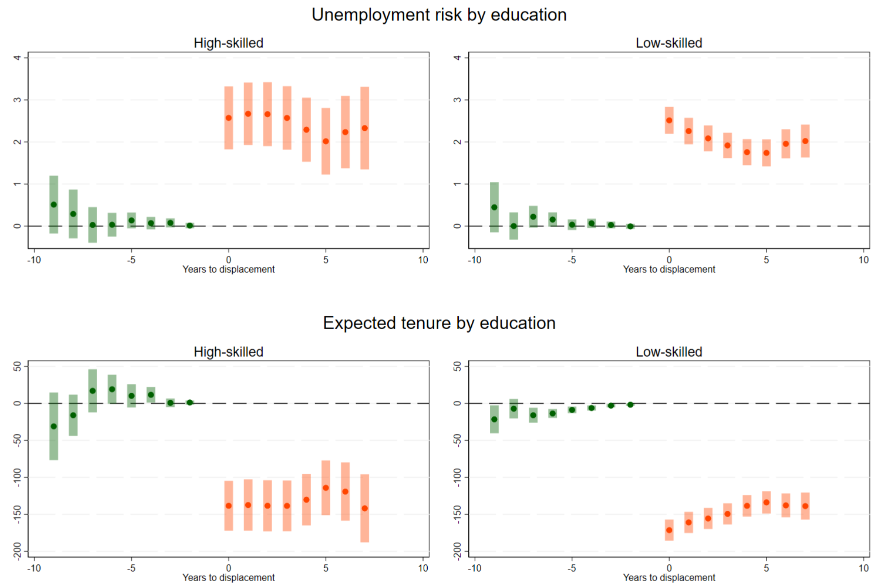
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. The charts on the left show estimates obtained for individuals who join their next main occupation following dismissal after a period of six months. The charts' estimates on the right refer to individuals who joined their next main occupation within a time period of six months. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B10: Conditional regression à la Callaway and Sant'Anna (2021) by gender - control group: never-treated units



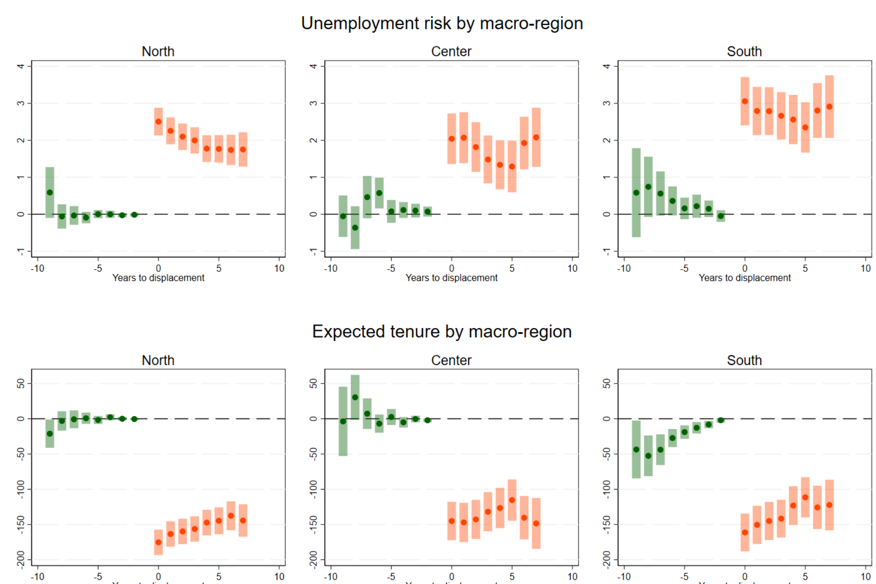
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B11: Conditional regression à la Callaway and Sant'Anna (2021) by educational level - control group: never-treated units



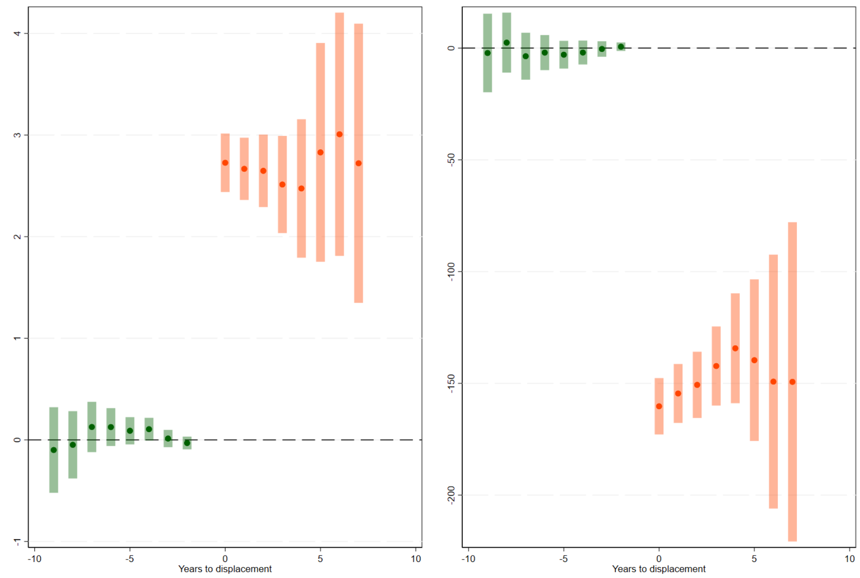
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B12: Conditional regression à la Callaway and Sant'Anna (2021) by macro-region of work - control group: never-treated units



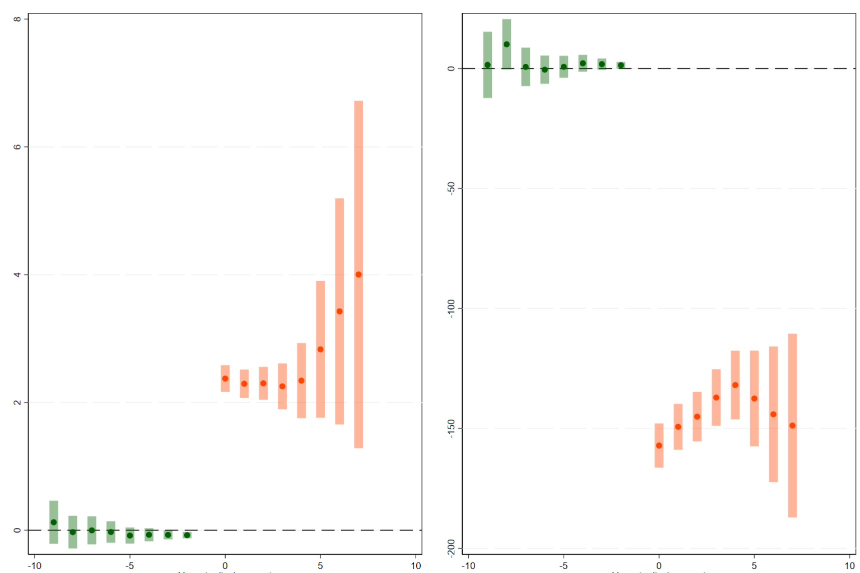
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (top) and expected tenure (bottom) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using never-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B13: Sensitivity: excluding outlier observations in the bottom and top 1% of the outcome variables' distributions



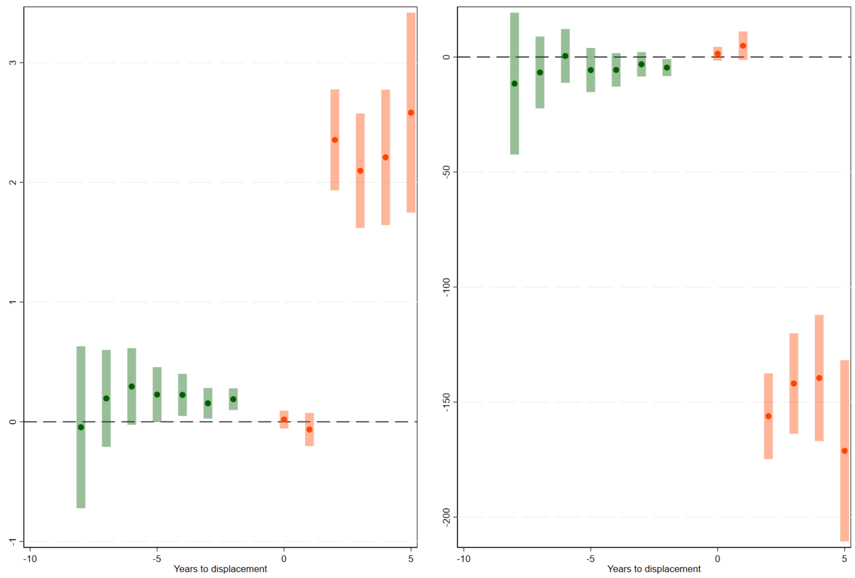
Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B14: Conditional regression à la Callaway and Sant'Anna (2021) - robustness: including individuals aged 20 to 64



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.

Figure B15: Conditional regression à la Callaway and Sant'Anna (2021) - placebo: treatment reallocated two years prior to the real one



Note: The chart displays estimates of the ATT of displacement, expressed as the variation in unemployment risk (left) and expected tenure (right) associated with the main occupations in which "treated" individuals are employed each year. Estimates are obtained using not-yet-treated units as a control group. Confidence intervals, represented by the extent of the bars, are computed using a 95% confidence level.